

California Wildfire Analysis

Trends, Damage Patterns, and Building Feature Risk Factors

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Data Visualization (ISM6361)

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April 30, 2025

Business Understanding

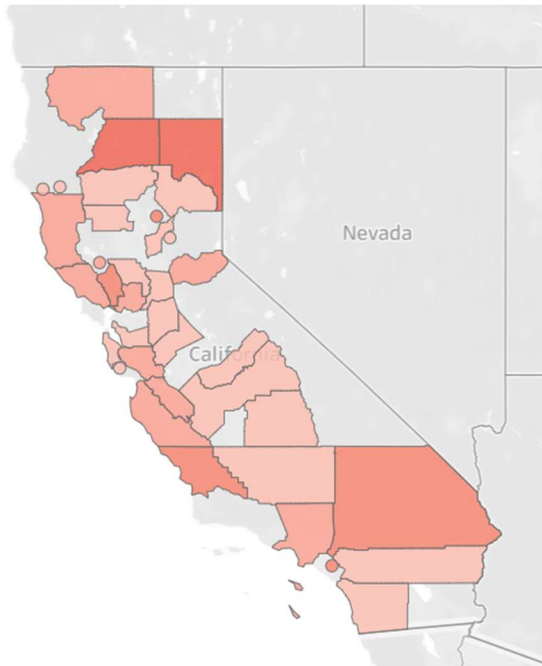
Wildfires in California have been significantly increasing in intensity and size due to a combination of climate change, heat waves, prolonged dry conditions, and continued human-induced activities, greatly affecting the economy over the past few years. As more houses and other infrastructures are built in areas with a high risk of wildfire, wildfire events are causing greater destruction to property and ecosystems, loss of life, and costs for emergency response and recovery.

In extreme cases like the 2020 wildfire season, as shown in Figure 1, a record-breaking heat, widespread dry lightning storms, and strong winds led to the burning of millions of acres across California, proving how environmental and human factors are creating unprecedented fire conditions. Moreover, fire insurance companies declined to renew 2.8 million homeowner policies in California between 2020 and 2022 due to these wildfire disasters, leaving people with significant disruption and financial burdens. (Isidore, 2025)

This report analyses the major trends in California wildfire incidents from 2013 to 2024, focusing on how wildfire frequency and severity have changed over time, how property exposure has evolved, and how specific building features influenced a structure's vulnerability to wildfire damage. By identifying time-based trends and structural risk factors, it aims to provide recommendations for improving community resilience, guiding prevention funding, and promoting building practices that reduce wildfire losses. A deeper understanding of these patterns can be used to protect lives, homes, and ecosystems as wildfire threats continue to escalate across California.

Figure 1

Geographic distribution of major wildfire incidents across California in 2020



Note. This figure shows the widespread geographic reach of wildfire in 2020, one of the most destructive in recent history.

Data Description and Summary

This analysis report uses a California Post-Fire Parcel Inspection Dataset with real post-fire structure assessments from 2013 to 2024, which are inside or within 100 meters of the fire perimeter. Each record corresponds to a single parcel inspected for damage and provides detailed information about building characteristics and fire impacts, holding 100,230 structure assessments.

Key variables include incident start dates, assessed improved property value, and damage severity classifications, along with building features such as roof type, exterior siding material, eaves construction, and vent screen presence. Damage severity is categorized into levels ranging

from “No Damage” to “Destroyed”, allowing for analysis of survival rates across different structure types. Moreover, records with missing or invalid entries for essential fields were removed to ensure data quality.

As part of preprocessing, the building features were grouped into meaningful categories, and a Damage Rate field was created to calculate the percentage of structures affected at each severity level. Also, assessed property values were binned into logical ranges to support clearer comparisons across different building value levels.

Because this dataset only includes wildfires where recorded structure inspections were conducted, it does not capture every wildfire. As a result, it does not reflect every fire reported, focusing specifically on major fire events where damage or other structures were assessed.

Business Questions

This report answers five primary business questions related to the patterns, severity, and structural impacts of California wildfires:

1. How have wildfire impacts evolved across California over time?
2. How has wildfire frequency and the total value of exposed properties changed from 2013 to 2024?
3. What months of the year are wildfires most likely to occur, and how has the fire season shifted?
4. How does the assessed property value relate to the severity of wildfire damage experienced by structures?
5. How do specific building features such as roof type, exterior siding, eaves construction, and vent screening affect a structure’s vulnerability to wildfire damage?

These questions were developed to explore the changing nature of wildfire incidents and the practical factors that influence the survivability of homes and infrastructure when exposed to fire. Each question is addressed through geographic analysis, time-based trend evaluation, structural feature examination, and property value comparisons.

Findings

1. How have wildfire impacts evolved across California over time?

Wildfire impacts have grown dramatically across California between 2013 and 2024. Early in this period, wildfire activity was mostly concentrated in parts of Northern California, such as Shasta and Butte Counties, and in parts of Southern California, such as Ventura County. However, fires became more geographically widespread in 2017, moving into areas that historically had fewer fire events. By 2020, wildfire incidents were recorded across nearly the entire state, fueled by a combination of record-breaking heat, prolonged drought, massive lightning storms, and stronger wind events that made fire spread uncontrollably. (Figure 2)

It also reached more densely populated areas, threatening suburban communities and areas closer to major urban centers. These findings show that wildfire threats are no longer limited to remote or rural areas, demanding greater needs on fire codes, emergency planning, and public education efforts across all California regions.

Figure 2

Animated map of wildfire expansion across California from 2013 to 2024 (Please view the video on the Word document)

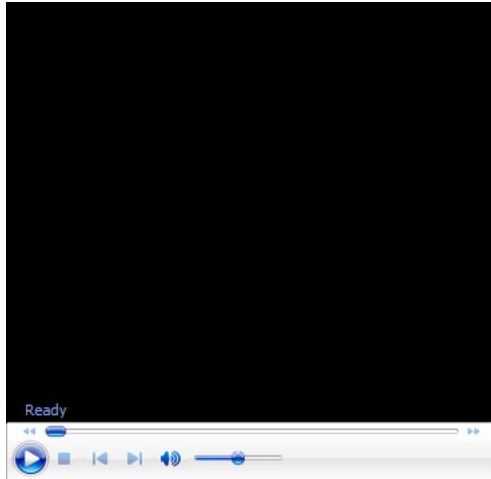


 Figure 2.mp4

2. How has wildfire frequency and the total value of exposed properties changed from 2013 to 2024?

Figure 3 shows that the number of wildfire incidents fluctuated each year, while the total value of property exposed to wildfire risk rose gradually during the period analyzed. In the early years, fires mainly affected lower-value rural properties. However, by 2018 and 2020, fire activity increasingly occurred near suburban neighborhoods and high-value residential developments. The 2020 season was particularly significant, with billions of dollars in property value exposed due to fires burning closer to larger cities and denser communities. Moreover, in years with few total fire events, the financial risk continued to grow from wildfires increasingly threatening areas with higher concentrations of property wealth.

These findings indicate that the risk of wildfires cannot be measured by fire counts alone, and preventive efforts must prioritize areas with higher property exposure, even if the number of annual ignitions appears lower. In addition, programs focused on defensible space and structure, and evacuation planning in these vulnerable zones can reduce catastrophic property loss.

Figure 3

Wildfire Growth and Impact on Property: 2013-2024



3. What months of the year are wildfires most likely to occur, and how has the fire season shifted?

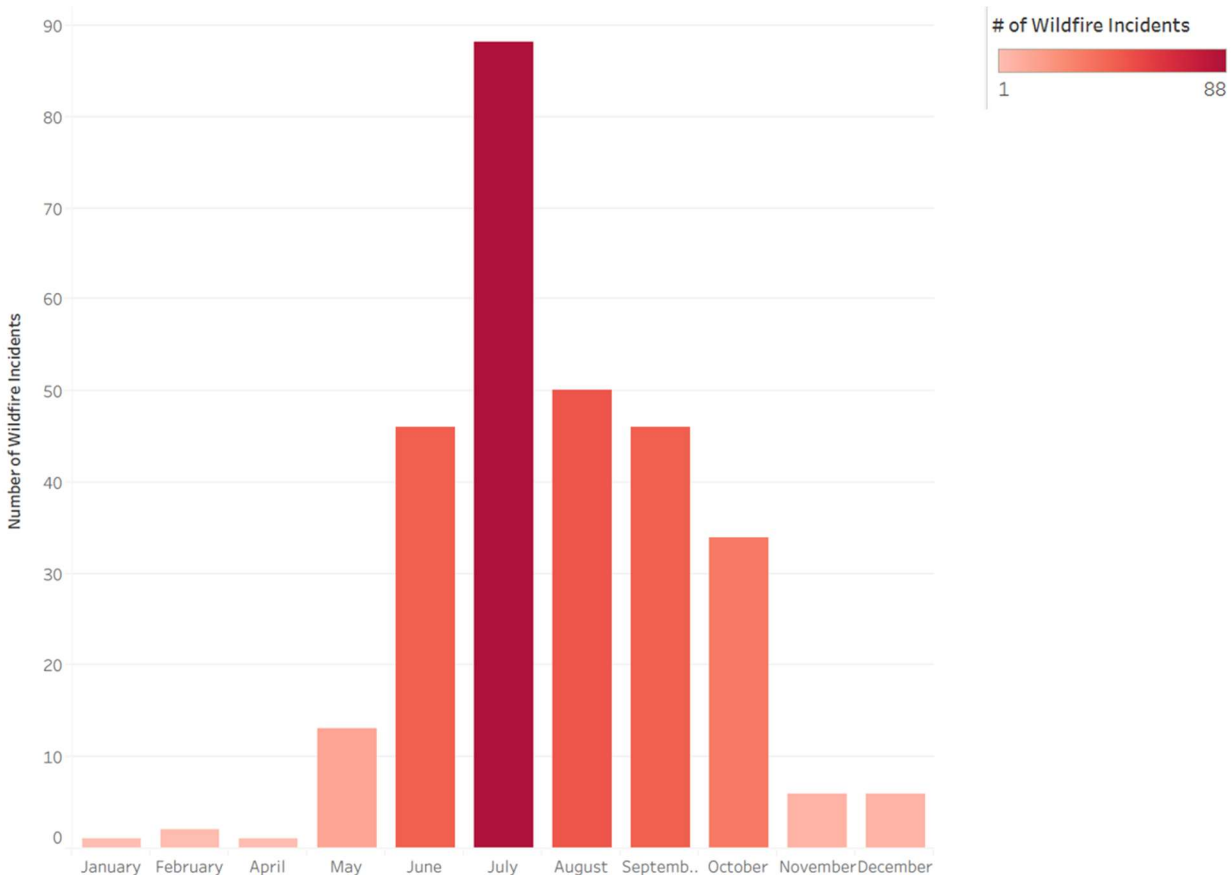
Historically, wildfires in California occurred mostly during late summer, primarily in July and August. However, Figure 4 shows that wildfire activity has expanded into a much longer season, with significantly higher numbers of fires from June through November, with November representing a major peak.

Fall fires have become particularly dangerous because dry fuels accumulated during the long summer are then ignited and rapidly spread by seasonal dry winds. These patterns confirm that California now faces a year-round fire management challenge, with no sharp endpoint after summer. Due to this, fire agencies and communities must adapt to the change by building fire preparedness activities, defensible space maintenance, and emergency readiness deeper into the fall months. Moreover, other critical solutions such as updating public alert systems and

extending seasonal firefighting staffing should also be applied to manage the new, longer, and more dangerous fire seasons.

Figure 4

Wildfire Frequency by Month



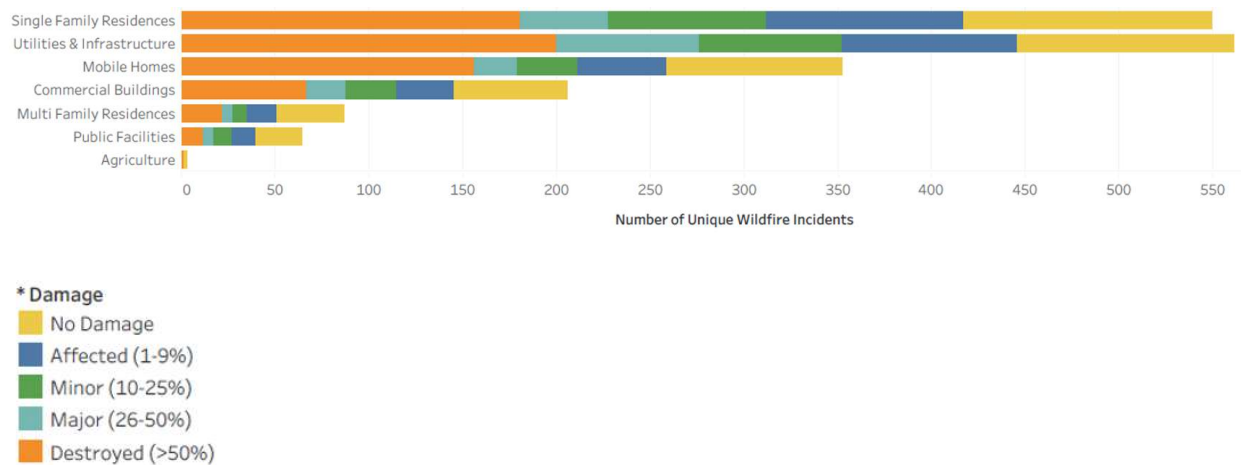
4. How does the assessed property value relate to the severity of wildfire damage experienced by structures?

Figure 5 shows that homes with higher assessed property values were more likely to survive wildfire events compared to mid-value and lower-value homes. Higher-value properties often included better-quality building materials, upgraded fire-resistant construction, professionally managed defensible space, and private fire protection services.

However, wildfire destruction occurred across all property tiers, and lower-value and mid-value homes experienced much higher rates of total destruction. This reflects disparities in access to fire mitigation resources and the construction standards applied to different neighborhoods, underlining the need for equitable fire mitigation programs. Policies should prioritize grants, subsidies, and community support services that help mid- and lower-income homeowners implement critical fire-hardening upgrades.

Figure 5

Wildfire Impact by Structure Type and Damage Severity



5. How do specific building features such as roof type, exterior siding, eaves construction, and vent screening affect a structure’s vulnerability to wildfire damage?

Building features play a critical role in determining whether a home or structure survives wildfire exposure. Analysis from Figure 6 showed that structures with fire-resistant roofs, such as metal and tile, had much higher survival rates than those with wood shake or other combustible roofing materials, and exterior wall material also strongly influenced survival.

Moreover, structures built with stucco or brick exteriors performed significantly better compared to those with vinyl siding or untreated wood siding, both of which are more easily

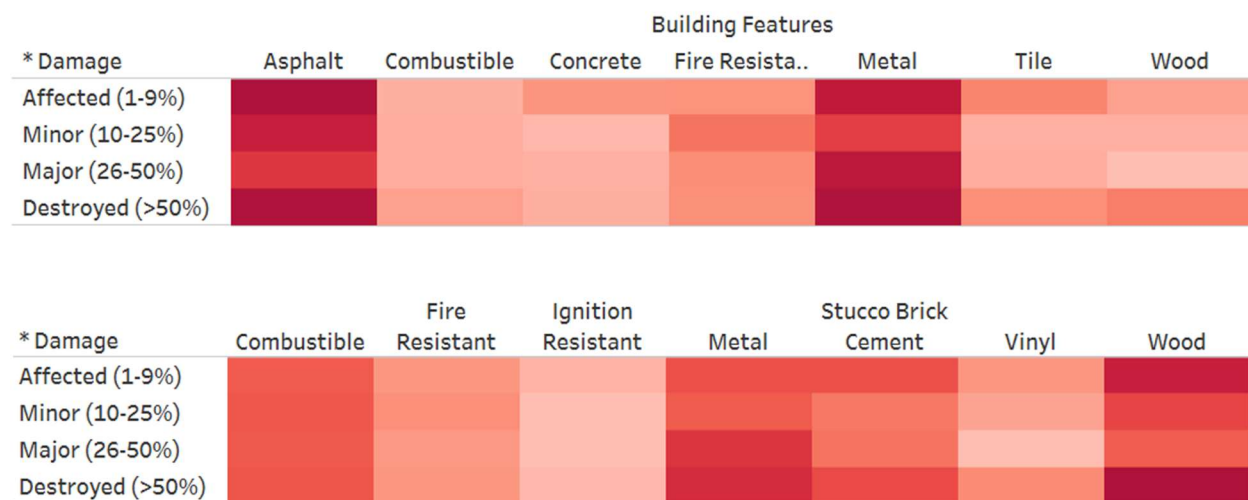
ignited by embers or radiant heat. Additionally, structures with enclosed eaves had substantially lower destruction rates than those with open or exposed eaves, as enclosed eaves prevented embers from entering attic spaces.

Finally, homes with fine mesh vent screens (one-eighth-inch openings or smaller) showed far lower internal ignition rates compared to homes with no vent screens or larger mesh openings. These findings demonstrate that relatively small construction choices make an enormous difference in wildfire survival rate.

Solutions such as mandating enclosed eaves, installing ember-resistant vents, promoting the use of fire-resistant roofing and siding materials, and retrofitting existing structures with these features can significantly reduce wildfire destruction rates statewide. Although cost is often the primary barrier preventing widespread adoption of these, targeted funding, tax deduction incentives, and public education programs can help make these improvements more accessible and achievable for homeowners across income levels.

Figure 6

Wildfire Damage Rates by Building Feature Type



* Damage	Enclosed	No Eaves	Unenclosed
Affected (1-9%)			
Minor (10-25%)			
Major (26-50%)			
Destroyed (>50%)			

* Damage	Mesh Screen <= 1/8'''	Mesh Screen > 1/8'''	No Vents	Screened	Unscreened
Affected (1-9%)					
Minor (10-25%)					
Major (26-50%)					
Destroyed (>50%)					

Business Recommendations

One primary recommendation from this analysis is to strengthen fire-resistant building standards statewide. The findings clearly demonstrate that construction features dramatically improve a structure's chances of surviving wildfire exposure. Based on this evidence, California should expand existing building code requirements for new construction to mandate these features in all wildfire-prone areas.

In addition, public initiatives should provide financial incentives and retrofitting programs that help existing homeowners fix older structures. These efforts could be supported by funding the development of practical solution tools, such as home risk assessment apps or wildfire resilience planning platforms, to help residents understand their vulnerabilities and make informed decisions about mitigation. Strengthening construction standards will directly reduce property losses, improve community resilience, and lower the human and economic costs associated with future wildfire events.

Another recommendation is to prioritize wildfire prevention resources toward mid-value and high-risk residential areas, rather than focusing exclusively on high-income regions. The analysis revealed that mid-value and lower-value properties often experienced disproportionately higher destruction rates because they are less likely to have fire-resistant features and maintained defensible spaces. Therefore, prevention funding should be directed toward these targeted neighborhoods to not only save more lives and reduce overall property losses, but also to create a stronger and more equitable foundation for community recovery after future fires.

Finally, public awareness campaigns should be expanded to promote simple, affordable home-hardening techniques that homeowners can implement immediately. Educational efforts should emphasize that critical updates, such as installing fine mesh vent screens, enclosing eaves, cleaning roofs and gutters, and removing flammable materials near structures, can significantly improve a home's ability to withstand wildfire conditions.

Outreach should particularly target homeowners living at the expanding edges of cities and suburbs, where fire activity is growing but building codes have often not yet caught up. By empowering more residents with knowledge about ember resistance and small construction improvements, communities can achieve faster increases in wildfire resilience without relying solely on government intervention.

Faith and Ethics Implications

Wildfire risk management is not only a technical challenge, but also an ethical responsibility. The findings of this report show that the consequences of wildfires are not distributed equally across communities. From an ethical standpoint, wildfire prevention should be guided by principles of fairness, stewardship, and compassion.

Resources such as grants for retrofitting homes, community fire-hardening initiatives, and emergency response planning should prioritize areas that have historically been under-protected or that lack fire resilience measures independently. These programs should be structured to reward proactive fire-safe construction and discourage the continued use of high-risk, non-resistant materials chosen solely for their lower cost, ensuring that public funding supports long-term safety rather than short-term savings.

From a Christian faith perspective, there is a strong call to stewardship and neighborly love. Stewardship involves responsible care of the natural world, including making wise choices to protect homes, forests, and ecosystems from preventable destruction. Neighborly love calls for a commitment to care for others, particularly those who are vulnerable or at greater risk.

Applying these values to wildfire resilience means promoting not only personal property protection but also community-wide preparedness that does not leave lower-income or high-risk residents behind. This ethical wildfire management must emphasize the protection of all lives and livelihoods. By aligning fire prevention policies with values of justice, stewardship, and service to others, communities can build stronger, fairer, and more resilient futures in the face of increasing wildfire threats.

Conclusion

Wildfires have become a growing threat to communities across California, with impacts growing in intensity, scale, and cost over the past decade. This report's analysis shows that wildfire risks are no longer confined to remote forests but have extended into suburban and urban areas, threatening more homes, infrastructure, and lives. Although the number of wildfires fluctuates annually, the overall property value at risk continues to rise, driven by climate change,

prolonged droughts, and expanding development into fire-prone regions. These demonstrate that construction choices have a profound effect on a structure's ability to survive wildfire exposure.

By identifying these risk factors and connecting them to clear solutions, this report highlights the importance of proactive fire mitigation strategies at both the individual and community levels. Strengthening fire-resistant building standards, prioritizing prevention resources for vulnerable neighborhoods, and promoting affordable home-hardening techniques are key steps toward reducing future wildfire destruction. Furthermore, addressing wildfire resilience through stewardship, fairness, and service ensures that protection efforts are inclusive and just.

Lastly, building wildfire resilience is not only about preventing losses but about safeguarding communities, preserving ecosystems, and respecting the interconnected responsibilities we share. Through data-driven decision making, public education, ethical action, and government support, California can move toward a future where wildfire threats are met with stronger, smarter, and more equitable defenses.

References

- Aliferis, L. (2025, January 8). *California's worsening wildfires explained*. CapRadio.
<https://www.caprado.org/articles/2025/01/08/californias-worsening-wildfires-explained/>
- Brangham, W., Lane, S., & Fritz, M. (2025, February 10). *California faces insurance crisis as homeowners lose coverage amid extreme weather*. PBS NewsHour.
<https://www.pbs.org/newshour/show/california-faces-insurance-crisis-as-homeowners-lose-coverage-amid-extreme-weather>
- CAL FIRE Damage Inspection (DINS) data - CSV - California Open data. (n.d.).
<https://data.ca.gov/dataset/cal-fire-damage-inspection-dins-data/resource/b8aeb030-140d-43d2-aa29-1a80862e3d62>
- Chandler, J. (2024, May 20). *Why California's wildfires are getting worse*. UCLA Newsroom.
<https://newsroom.ucla.edu/releases/experts-explain-why-california-wildfires-getting-worse>
- Isidore, C. (2025, January 9). *California's insurance is in crisis. The solution will cost homeowners a ton*. CNN Business. <https://www.cnn.com/2025/01/09/business/california-wildfires-homeowners-insurance/index.html>
- National Integrated Drought Information System. (2023, August 8). *Study finds climate change to blame for record-breaking California wildfires*. Drought.gov.
<https://www.drought.gov/news/study-finds-climate-change-blame-record-breaking-california-wildfires-2023-08-08>

Trafton, A. (2025, January 15). *How and why the L.A. wildfires grew so fast*. Boston University.

<https://www.bu.edu/articles/2025/how-and-why-the-la-wildfires-grew-so-fast/>